



GB4007:Supplier Controller Provisioning Portal Manual

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Owner:	Manohar Reddy 248.953.9135 manohar.reddy@gm.com
Approved by:	Cybersecurity CCB 2435625, 2096644,1022628
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1 Introduction - GUID: CYS-SCPPM893c3_30

GUID: CYS-SCPPMa0c61_91 / CR 596107 (information only)

The intent of this document is to provide GM suppliers with the information necessary to securely access and manage the ECU_IDs and Secret Keys required to be provisioned into microcontrollers during the ECU manufacturing process.

Certain ECUs targeted for the Global B electrical architecture are required to be provisioned with an ECU_ID and one or more Secret Keys during the ECU manufacturing process. Although an ECU_ID is required to be provisioned into all ECUs that require any type of Secret Key, the specific Secret Key(s) to be provisioned into a given microcontroller are dependent on the microcontroller hardware.

Global B ECUs which are required to support Message Authentication must utilize a microcontroller with an integrated security peripheral. All instances of an integrated security peripheral must be in compliance with the General Motors Security Peripheral requirements as defined in CYS2300 - HSM Core Requirements (for ECUs with HSM based security peripheral) or CYS2200 - SHE+ Requirements (for ECUs with SHE+ based security peripheral). Alternately, some program teams have opted to include these same Security Peripheral requirements into a program specific CTR or CTS.

If a design is required to support a security peripheral the supplier is required to provision both the MASTER_ECU_KEY and UNLOCK_ECU_KEY into specified key slots of the microcontroller security peripheral during the ECU manufacturing process. If a design does not require support of a security peripheral but will support the GM UDS SecurityAccess service, the supplier will be required to provision the UNLOCK_ECU_KEY into secure non-volatile memory (secure static NVM) of the microcontroller during the ECU manufacturing process. Unlike the Secret Key dependency on microcontroller hardware, the ECU_ID shall always be provisioned into the One Time Programmable (OTP) memory of microcontrollers required to support it. If OTP is not available, then the ECU_ID shall be provisioned into the secure static non-volatile memory (NVM) of microcontrollers required to support it.

GUID: CYS-SCPPMb4d180_236

GM provides a matched set of ECU_ID and Secret Key(s) (i.e. MASTER_ECU_KEY and/or UNLOCK_ECU_KEY) via a GM Portal.

GUID: CYS-SCPPMe5f9c0_276 / CR 2096644

GM IECS Web Portal is accessible via <https://gmsupplypower.covisint.com/web/portal>

Note: Suppliers need to click on "IECS Portal" link from GM Supply Power =>COLLABORATE => Applications

GUID: CYS-SCPPM13d160_110 / CR 596107 (information only)

Access to a GM Portal is made available to suppliers for purposes of retrieving the ECU_ID and Secret Key(s) (i.e. MASTER_ECU_KEY and/or UNLOCK_ECU_KEY) to be provisioned into each ECU during ECU manufacturing. Each supplier will have remote access to the GM Portal. The Portal allows for the supplier to retrieve a requested number of provisioning data sets to allow the supplier to build the expected number of ECUs. Each Provisioning Data set includes the ECU_ID

and Secret Key(s) required for a given ECU. Each Provisioning Data set is unique to an ECU and may not be reused after provisioned into an ECU.

Upon a successful request, the GM Portal will upload a file containing the requested number of Provisioning Data sets to the users (supplier) computer. The file containing the requested Provisioning Data sets is referred to as the "Supplier Download File".

Standard internet security protocols will be employed to protect the transfer of the Supplier Download File between the Portal and supplier's computer.

Designated supplier manufacturing personnel must be on-boarded into GM Supply Power prior to obtaining access to the Portal. Once on-boarded into GM Supply Power, the user will be able to request access to the Portal Application. From this point forward, the supplier will work with the GM Security Administrator to configure the individual production needs of each ECU program requiring data from this system.

Note: The technical support contact information is available on the Portal web page.

GUID: CYS-SCPPMa0c61_88 / CR 596107 (information only)

The Supplier Download File retrieved from the Portal is protected using OpenPGP to ensure the file is secured when stored at rest on the supplier's computer. OpenPGP is also used to provide the supplier with a signature for purposes of ensuring GM is the source of the Supplier Download File. To facilitate in the use of OpenPGP the supplier will be tasked to provide GM with a Supplier Public Key to be used for encryption of the Supplier Download File as described in CYS-SCPPM13d160_112. GM will provide the supplier with a GM Public Key for purposes of verifying the signature on the Supplier Download File. During the onboarding process each supplier will be required to generate a public / private key pair for use in the PGP protocol. As part of the Supply Power on-boarding process a GM Security Administrator will be given notice of the supplier request for system access and will contact the specific user being configured to exchange GM and Supplier Public Keys. Each user must be affiliated with a Supplier Public Key but multiple users from the same supplier can share common public keys and it is at the discretion of the supplier to use a single public key per user or the same public key for multiple users.

The Supplier Public Key will be used to encrypt the Supplier Download File where the supplier must therefore use the Supplier Private Key to decrypt the Supplier Download File.

The GM Private Key will be used to sign the Supplier Download File where the supplier must therefore use the GM Public Key to verify the signature on the Supplier Download File.

GUID: CYS-SCPPMa0c61_92 (information only)

A complete definition of functionality associated with ECU_IDs and any Secret Keys is outside the scope of this document and can be found between the ECU Security Requirements section of the CTRS (or a CTR / CTS section derived from this material) and the following specifications:

- GB4001 - ECU Unlock Specification

- GB4002 - In Vehicle Message Authentication

- GB6002 - Bootloader Specification

GUID: CYS-SCPPMb4d180_237

GM provides the Secret Key(s) (i.e. MASTER_ECU_KEY and/or UNLOCK_ECU_KEY) for the sole purpose of provisioning them into the ECUs. GM will not provide these Secret Key(s) for any other purposes including debugging, remanufacturing, etc.

2 ECU_ID - GUID: CYS-SCPPM893c3_32

GUID: CYS-SCPPM893c3_33 (information only)

The ECU_ID is a unique 16-byte value derived for each instance of a microcontroller. Any design which supports SecurityAccess or Message Authentication will require the support of an ECU_ID. No two microcontrollers will have the same ECU_ID.

GUID: CYS-SCPPMa0c61_86

The ECU_ID shall be provisioned into One Time Programmable (OTP) memory, if available. If OTP is not available, then the ECU_ID shall be provisioned into the secure static non-volatile memory (NVM) of the microcontroller.

Note: Refer to CTRS for requirements on secure static NVM.

GUID: CYS-SCPPMb4d180_223

Each instance of a microcontroller communicating on a public bus shall be provisioned with unique ECU_ID as provided by GM.

GUID: CYS-SCPPMb4d180_224

Each microcontroller supporting secure programming shall be provisioned with an ECU_ID as provided by GM.

GUID: CYS-SCPPMb4d180_238

Regardless of the use case, no two ECUs ever delivered to GM shall have the same ECU_ID.

GUID: CYS-SCPPM13d160_111 / CR 596107 (parameter definition)

The ECU_ID is provided to the ECU supplier as a 16 byte binary value.

GUID: CYS-SCPPMb4d180_239

Production/fielded microcontrollers shall not contain any method to be able to provision, reprovision, change or erase the ECU_ID from the microcontrollers.

Note: An assumed exception to this is the JTAG/HW debug port.

3 Secret Keys - GUID: CYS-SCPPMa0c61_97

3.1 Microcontrollers without a Security Peripheral - GUID: CYS-SCPPM893c3_34

3.1.1 UNLOCK_ECU_KEY - GUID: CYS-SCPPM893c3_35

GUID: CYS-SCPPM893c3_36 (parameter definition)

For ECUs containing a microcontroller which does not support a security peripheral the UNLOCK_ECU_KEY is provided to the ECU supplier as a 16 byte binary value.

GUID: CYS-SCPPM893c3_75 (information only)

For ECUs containing a microcontroller which does not support a security peripheral (ISPKSP = 0x00 and NOKSS = 0x00) the ECU supplier is responsible for writing the UNLOCK_ECU_KEY to the required One Time Programmable (OTP) or secure static non-volatile memory (NVM) location within the microcontroller memory.

The ECU production mechanisms used to write the UNLOCK_ECU_KEY to microcontroller are left to the discretion of the supplier. These mechanisms are not allowed to be resident in a production/fielded ECU.

GUID: CYS-SCPPMa0c61_83 / CR 596107

If a design does not require support of a security peripheral (ISPKSP = 0x00 and NOKSS = 0x00) but will support the GM UDS SecurityAccess service, the supplier shall provision the GM provided UNLOCK_ECU_KEY into One Time Programmable (OTP) memory, if available. If OTP is not available, then the UNLOCK_ECU_KEY shall be provisioned into the secure static non-volatile memory (NVM) of the microcontroller.

Note: Refer to CTRS for requirements on secure static NVM.

GUID: CYS-SCPPMb4d180_240

GM provided UNLOCK_ECU_KEY shall be provisioned into the ECU during the ECU manufacturing process.

GUID: CYS-SCPPMb4d180_241

Supplier shall ensure that the UNLOCK_ECU_KEY provisioned into the ECU during the ECU manufacturing process is part of the matched set of ECU_ID and UNLOCK_ECU_KEY as provided by GM.

Note: For example, the supplier shall ensure that they only provision UNLOCK_ECU_KEY that matches the ECU_ID programmed into the controller.

GUID: CYS-SCPPMb4d180_242

Production/fielded ECUs shall not contain any method to be able to provision, reprovision, change or erase the UNLOCK_ECU_KEY from the ECU.

Note: An assumed exception to this is the JTAG/HW debug port.

GUID: CYS-SCPPMb4d180_243

Regardless of the ECU use case or the process via which the supplier received the UNLOCK_ECU_KEY, the UNLOCK_ECU_KEY shall be destroyed from all supplier IT systems after provisioning in the ECU.

Note: Refer to Third Party Information Security Requirements (TPISR) for definition and process details of 'data destruction'.

3.2 Microcontrollers with a Security Peripheral - GUID: CYS-SCPPM893c3_37

GUID: CYS-SCPPMb4d180_244 (information only)

GM provided the Secret Keys (i.e. MASTER_ECU_KEY and UNLOCK_ECU_KEY) cannot be erased from the microcontroller and cannot be reprovisioned within the same microcontroller. If any errors are detected while provisioning GM provided the Secret Keys, the part (microcontroller and/or ECU) will need to be scrapped.

GUID: CYS-SCPPMb4d180_225

For ECUs containing a microcontroller which does support a security peripheral (ISPKSP = 0x01 or NOKSS > 0x00), MASTER_ECU_KEY and UNLOCK_ECU_KEY shall be initialized/provisioned to zero key (16 bytes of zeros) before provisioning GM provided MASTER_ECU_KEY and UNLOCK_ECU_KEY.

GUID: CYS-SCPPMb4d180_226

Supplier shall generate M1, M2 and M3 for provisioning zero keys (16 bytes of zeros) in any key slot. To ensure that zero key provisioning was successful suppliers are recommended to check M4 and M5 returned by the security peripheral.

Note: GM will not provide M1,M2...M5 for provisioning zero keys(16 bytes of zeros) for any key slot. Refer to Microcontroller/Microprocessor user manual on how to provision zero key.

3.2.1 MASTER_ECU_KEY - GUID: CYS-SCPPM893c3_40

GUID: CYS-SCPPM893c3_41 (parameter definition)

For ECUs containing a microcontroller which does support a security peripheral (ISPKSP = 0x01 or NOKSS > 0x00), MASTER_ECU_KEY will be provided by GM as a set of M1, M2, M3 messages compliant with the security peripheral memory update protocol. The M1 message is 16 bytes in length, the M2 message is 32 bytes in length, and the M3 message is 16 bytes in length. The set of M1, M2, M3 messages provided for the MASTER_ECU_KEY is represented in the binary format.

GUID: CYS-SCPPM893c3_77 / CR 1022628

The ECU supplier shall load the GM provided MASTER_ECU_KEY into the security peripheral (ISPKSP = 0x01 or NOKSS > 0x00) by passing the M1, M2, M3 messages to the security peripheral using the specific security peripheral interface for loading keys.

Note: The production mechanism of providing the M1, M2, M3 messages to the security peripheral interface is left to the discretion of the supplier. Diagnostics RID 0x0272 is available to provision any security peripheral keys and read back M4/M5 for verification.

GUID: CYS-SCPPM46b9e0_208 / CR 1007259

Regardless of the ECU use case or the process via which the supplier received the MASTER_ECU_KEY, M1, M2 and M3 associated with MASTER_ECU_KEY shall be destroyed from all supplier IT systems after verification by Local Verification Utility.

Note: Refer to Third Party Information Security Requirements (TPISR) for definition and process details of 'data destruction'.

3.2.2 UNLOCK_ECU_KEY - GUID: CYS-SCPPM893c3_38

GUID: CYS-SCPPM893c3_39 (parameter definition)

For ECUs containing a microcontroller which does support a security peripheral (ISPKSP = 0x01 or NOKSS > 0x00), UNLOCK_ECU_KEY will be provided by GM as a set of M1, M2, M3 messages compliant with the security peripheral memory update protocol. The M1 message is 16 bytes in length, the M2 message is 32 bytes in length, and the M3 message is 16 bytes in length. The set of M1, M2, M3 messages provided for the UNLOCK_ECU_KEY is represented in the binary format.

GUID: CYS-SCPPM893c3_76 / CR 1022628

The ECU supplier shall load the GM provided UNLOCK_ECU_KEY into the security peripheral (ISPKSP = 0x01 or NOKSS > 0x00) by passing the M1, M2, M3 messages to the security peripheral using the specific security peripheral interface for loading keys.

Note: The production mechanism of providing the M1, M2, M3 messages to the security peripheral interface is left to the discretion of the supplier. Diagnostics RID 0x0272 is available to provision any security peripheral keys and read back M4/M5 for verification.

GUID: CYS-SCPPM46b9e0_209 / CR 1007259

Regardless of the ECU use case or the process via which the supplier received the UNLOCK_ECU_KEY, M1, M2 and M3 associated with UNLOCK_ECU_KEY shall be destroyed from all supplier IT systems after verification by Local Verification Utility.

Note: Refer to Third Party Information Security Requirements (TPISR) for definition and process details of 'data destruction'.

3.3 ECU Security Peripheral Verification Messages - GUID: CYS-SCPPM893c3_42

GUID: CYS-SCPPM893c3_43 / CR 596107 (information only)

As part of the security peripheral design, verification message M4 and M5 will be provided by the security peripheral immediately following a successful key update. Suppliers responsible for ECUs which have a Security Peripheral are required to provide the M4 and M5 messages resulting from loading the MASTER_ECU_KEY and UNLOCK_ECU_KEY to GM.

GUID: CYS-SCPPM893c3_80 / CR 596107

The M4 and M5 verification messages resulting from loading the MASTER_ECU_KEY and UNLOCK_ECU_KEY shall be provided back to GM through the available Provisioning Portal Verification Interface prior to the ECU arriving to GM facilities.

GUID: CYS-SCPPMb4d180_245 / CR 2435625

ECUs that fail verification through the GM Portal (verification result report shows ECU is failed) shall be destroyed.

Note: ECUs that fail verification checks must be blacklisted from IECS system. Suppliers to keep track the list of blacklisted ECUIDs and send it to cys@gm.com once in every 90 days. List of ECUIDs should be in readable hex format.

GUID: CYS-SCPPM893c3_78 / CR 596107 (information only)

The GM Portal enforces the requirement of providing M4 and M5 verification messages back to GM by limiting suppliers the ability to retrieve future sets of Provisioning Data once a defined “number of unverified M4 and M5 messages” is reached. For each new Provisioning Data set provided by the Portal, the “number of unverified M4/M5 messages” is incremented by one. For each combined set of M4/M5 resulting from loading both the MASTER_ECU_KEY and UNLOCK_ECU_KEY provided to and successfully verified by the Portal, the “number of unverified M4/M5 messages” is decremented by 1. These verification messages do not need to be provided to GM immediately after the security peripheral provides them, rather they must be received by GM before the ECU arrives to a GM location.

The GM Portal also restricts the number of Provisioning Data sets permitted over a defined period of time. The number of Provisioning Data sets permitted is configured during the onboarding process and is based on the expected volume.

Note: If the supplier is made aware of production volume changes that will increase the number of keys needed, it is the responsibility of the supplier to request additional keys.

For example, suppose that a supplier has established the need for 100 sets of provisioning data and the supplier obtains all 100 for that period. At some point, the supplier uses 10 sets and then provides 10 sets of M4/M5 – portal would allow supplier to obtain only 10 sets of provisioning data in the next period. The supplier can then send 20 more sets of M4/M5 and would then be permitted to obtain another 20 sets of provisioning data. Once the production limit of 100 is reached for that period, they will no longer be able to obtain provisioning data until the next period.

Similarly, the supplier is limited to 100 sets of provisioning data for a given period. If supplier obtains 52 sets of provisioning data, they can still obtain 48 sets before any M4/M5s are required by Portal for that given period.

GUID: CYS-SCPPMb4d180_246

Supplier shall not provision the ECU_ID and the Secret Keys (i.e. MASTER_ECU_KEY and UNLOCK_ECU_KEY) that failed verification, either through the Local Verification Utility or through the GM Portal, into another ECU/microcontroller.

GUID: CYS-SCPPM12fc2e0_245 / CR 2435625

HSM ECU(s) with external flash shall commit Master Key and Unlock Key immediately before responding {M4, M5} back to host application.

4 Supplier ECU Provisioning Portal Web Interface - GUID: CYS-SCPPM893c3_44

GUID: CYS-SCPPM893c3_45 (information only)

A web based interface is available to allow suppliers remote access to retrieve the Supplier Download File. The interface will require user authentication to permit use of this service. Each supplier will be given a set of credentials to use when accessing the Supplier ECU Provisioning Portal. GM Supply Power is used for issuing of credentials to be used.

GUID: CYS-SCPPM893c3_86 / CR 2096644

Prior to the completion of the supplier onboarding process, the supplier shall create a public / private key pair with the public key being used to encrypt all batch file data. The GM portal security administrator will request the supplier public key during the onboarding process.

GUID: CYS-SCPPMb4d180_227 (information only)

Based on Electrical Architecture and Vehicle region of Sale, some sections from this chapter may not be applicable. See table below for details:

Note: RoW – Rest of the World including GMNA, GMSA, etc. For Global B, there is no EMPN proliferation.

Electrical Architecture	Vehicle Region of Sale/End Model Part Number(EMPN)	Applicable sections from this chapter	Contact information for IECS support
GEM	RoW	4.1, 4.2, 4.3	cys@gm.com
GEM	China	4.1, 4.2, 4.4	Cybersecurity PM
GB	RoW	4.1, 4.2, 4.3	cys@gm.com
GB	China	4.1, 4.2, 4.3	cys@gm.com

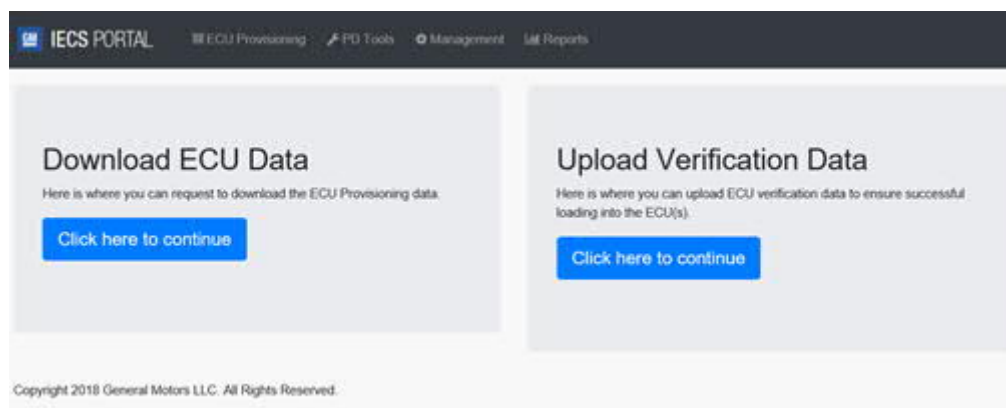
4.1 ECU Provisioning Data Interface - GUID: CYS-SCPPM893c3_48

GUID: CYS-SCPPM893c3_49 / CR 596107 (information only)

The user will be expected to select their ECU Product Name, the Number of ECUs to provide provisioning data for, and a local directory where the Supplier Download File is to be stored on the supplier's computer.

The ECU Product Name is pre-configured in the Portal by a GM Security Administrator.

Once logged into the ECU Provisioning Portal, the following screen will be presented. For purposes of requesting ECU provisioning data, the user will need to click the “Download ECU Data” button shown on the left pane of the screenshot below.



After selecting the “Download ECU Data” button, the following screen will be presented.



In this screen, user can either create a new request or download the generated file for previous requests.

After selecting the “Create New Request” button, the following screen will be presented. The user will need to choose the ECU Product of interest from the available “ECU Product Name” dropdown menu. The user will only see ECU Products which he/she has responsibility of. The user will then need to enter the number of ECU Provisioning Data sets required (i.e. the number of ECUs requiring provisioning). GM monitors and manages the amount of ECU Provisioning Data sets obtained for a given product lifetime and over a 28 rolling period.

It is important that the supplier pay particular attention to the ECU Provisioning Data Limit information included on the web page illustrated below. The ECU Provisioning Data Limit information includes:

- The max possible number of ECU Provisioning sets permitted over a 28 day rolling period
- The current number of ECU Provisioning sets retrieved over the most recent 28 day rolling period
 - o Once this value is equal to the max possible number of ECU Provisioning sets permitted over a 28 day rolling period, the Supplier ECU Provisioning Portal will not permit retrieval of ECU Provisioning Data sets pertaining to the ECU Product.
- The max possible number of ECU Provisioning sets which have not been verified using the Verification Data Interface described in CYS-SCPPM893c3_63
- The current number of ECU Provisioning Sets which have not been verified using the Verification Data Interface described in CYS-SCPPM893c3_63

- o Once this value is equal to the max possible number of ECU Provisioning sets which have not been verified, the Supplier ECU Provisioning Portal will not permit retrieval of ECU Provisioning Data sets pertaining to the ECU Product.
- The total number of ECU Provisioning sets retrieved over the life of the ECU Product

IECS PORTAL
ECU Provisioning
PD Tools
Management
Reports

Create ECU Provisioning Requests

ECU Product Name i AHL_AFL_Micro1_Confi v

Supplier Name i Continental

Supplier ID i 46

UPC/FNA i 12C/9770A

Supplier Appliance i DSABLZXBWMS

App Gen Key ID i 112

Security Peripheral? i true

ECU Provisioning Data Limits		
Data	Current	Maximum
Unverified Sets of ECU Provisioning Data	200	10000
Sets of ECU Provisioning Data Provided Over 28 days	0	100000
Total Sets of ECU Provisioning Data Provided	200	

of ECUIDs Needed i Number of ECU IDs

Percentage of ECUIDs unverified

Reset
Submit

Instructions:

To view Pending Requests go to Download ECU Dashboard

The following steps are required to make a download request:

1. Select a product from the "ECU Product Name" drop down list.
 - This will auto-populate the subsequent fields from the registration process: Supplier Name, Supplier ID, UPC/FNA, Supplier Appliance, App Gen Key ID, Security Peripheral)
2. Enter the number of ECU provisioning data sets needed for production or development.
3. Click "Submit" button.
4. Once your request is complete you will be notified via email. Go to the "Download ECU Dashboard" page to download the requested ECUIDs

Note: "Reset" will clear all fields.

Verify the file:

Please import our public key to your PGP software. This allows you verify the file we send by checking the signature.

View our public key

After selecting the "submit" button from the screen above, the user will get confirmation screen.

The screenshot shows the IECS PORTAL interface. At the top, there is a navigation bar with links for ECU Provisioning, PD Tools, Management, and Reports. A green notification banner states: "Your request has been submitted with the requestId=719EDEEA6B14C2B2E0539D12790A7D58. You will receive an email notifying you when the request is completed and ready for download." Below this, the "Download ECU Dashboard" is displayed, featuring a "Create New Request" button and a "Refresh" button. A message indicates that all file downloads will be available for 48 hours. A table lists the download requests with columns for Request ID, Supplier, Product, # of ECU IDs, GMIN, Request Date, Completed Date, Status, and Download. The first entry shows a request for 2 ECU IDs from WCM_Micro1_LGE, completed on 10-Aug-2018, with a status of "READY TO DOWNLOAD" and a download link icon.

Request ID	Supplier	Product	# of ECU IDs	GMIN	Request Date	Completed Date	Status	Download
719EDEEA6B14C2B2E0539D12790A7D58	Name	WCM_Micro1_LGE	2	551675292	10-Aug-2018 02:55 PM	10-Aug-2018 02:55 PM	READY TO DOWNLOAD	

Note: If user does not see the link in the “Download” column, select “Refresh” button.

Note: This download link is available only for 7 days. User needs to download this file before the link expires.

The file which is provided by the Supplier ECU Provisioning Portal is defined in CYS-SCPPM893c3_50.

GUID: CYS-SCPPM334b60_208 / CR 2435625 (information only)

If the supplier cannot download the ECU Data file, supplier must notify the project specific GM Product Cybersecurity Program Manager and copy email to cys@gm.com of the issue and request more keys to be generated or configured. ECU teams will be required to provide information on how and why the corruption occurred and what supplier will be doing to make sure that it does not occur again.

4.1.1 Output from ECU Provisioning Data Interface - GUID: CYS-SCPPM893c3_50

GUID: CYS-SCPPM893c3_51 / CR 596107 (information only)

The Supplier ECU Provisioning Portal will provide a binary file (Supplier Download File) containing ECU_IDs and Secret Keys for the number of ECUs selected. This binary file is encrypted with the user specific Supplier Public Key and will be downloaded to the user's computer into the directory selected by the user.

GUID: CYS-SCPPM893c3_82 (information only)

It is very important that the pairing of ECU_IDs and Secret Keys be provisioned together into the same ECU. A mismatch of the ECU_ID and Secret Keys results in an ECU which is not usable by GM.

GUID: CYS-SCPPMa0c61_87 / CR 596107

A single set of provisioning data shall only be used on one instance of an ECU (i.e. no two ECUs may be provisioned with the same set of ECU Provisioning Data).

GUID: CYS-SCPPMa0c61_90 / CR 596107 (information only)

A lost Supplier Download File cannot be retrieved and will count against the number of provisioning data sets retrieved.

4.1.1.1 Output for ECUs with a Microcontroller which does not have a Security Peripheral - GUID: CYS-SCPPM893c3_52

GUID: CYS-SCPPM893c3_53 (information only)

The Supplier Download File provided for ECUs with a Microcontroller which does not support a security peripheral will contain a pair of ECU_ID and UNLOCK_ECU_KEY for each device to be manufactured.

GUID: CYS-SCPPM893c3_54 (parameter definition)

The ECU_ID values for designs with a microcontroller which does not support a security peripheral will be provided in the Supplier Download File as 16 byte binary values represented in big endian byte order.

GUID: CYS-SCPPMa0c61_96 (parameter definition)

The UNLOCK_ECU_KEY values for designs with a microcontroller which does not support a security peripheral will be provided in the Supplier Download File as 16 byte binary values represented in big endian byte order.

GUID: CYS-SCPPMa0c61_94 (parameter definition)

The Supplier Download File structure for designs with a microcontroller which does not support a security peripheral is as follows:

```
ECU_ID_1
UNLOCK_ECU_KEY_1
...
...
ECU_ID_n
UNLOCK_ECU_KEY_n
```

4.1.1.2 Output for ECUs with a Microcontroller which has a Security Peripheral - GUID: CYS-SCPPM893c3_57

GUID: CYS-SCPPM893c3_58 (information only)

The Supplier Download File provided for designs with microcontrollers which support a security peripheral will contain a set of ECU_ID, MASTER_ECU_KEY, and UNLOCK_ECU_KEY values for each device to be manufactured.

GUID: CYS-SCPPM893c3_59 (parameter definition)

The ECU_ID values for designs with microcontrollers which support a security peripheral will be provided in the Supplier Download File as a 16 byte binary value represented in big endian byte order.

GUID: CYS-SCPPM893c3_60 (parameter definition)

The MASTER_ECU_KEY for designs with microcontrollers which support a security peripheral will be provided in the Supplier Download File as a set of message referred to M1, M2, and M3.

M1 for the MASTER_ECU_KEY will be provided as a 16 byte binary value represented in big endian byte order.

M2 for the MASTER_ECU_KEY will be provided as a 32 byte binary value represented in big endian byte order.

M3 for the MASTER_ECU_KEY will be provided as a 16 byte binary value represented in big endian byte order.

GUID: CYS-SCPPM893c3_61 (parameter definition)

The UNLOCK_ECU_KEY for designs with microcontrollers which support a security peripheral will be provided in the Supplier Download File as a set of message referred to M1, M2, and M3.

M1 for the UNLOCK_ECU_KEY will be provided as a 16 byte binary value represented in big endian byte order.

M2 for the UNLOCK_ECU_KEY will be provided as a 32 byte binary value represented in big endian byte order.

M3 for the UNLOCK_ECU_KEY will be provided as a 16 byte binary value represented in big endian byte order.

GUID: CYS-SCPPMa0c61_95 (parameter definition)

The Supplier Download File format for designs with a microcontroller that supports a security peripheral is as follows:

ECU_ID_1

MASTER_ECU_KEY_1_M1

MASTER_ECU_KEY_1_M2

MASTER_ECU_KEY_1_M3

UNLOCK_ECU_KEY_1_M1

UNLOCK_ECU_KEY_1_M2

UNLOCK_ECU_KEY_1_M3

...

...

ECU_ID_n

MASTER_ECU_KEY_n_M1

MASTER_ECU_KEY_n_M2

MASTER_ECU_KEY_n_M3

UNLOCK_ECU_KEY_n_M1

UNLOCK_ECU_KEY_n_M2

UNLOCK_ECU_KEY_n_M3

4.1.1.3 Supplier Download File Encryption/Decryption & Verification using OpenPGP - GUID: CYS-SCPPM13d160_112

GUID: CYS-SCPPMa0c61_89 / CR 596107 (information only)

The OpenPGP standard describes an efficient way to protect messages for communication over untrusted channels, such as the internet. GM's intent of using the OpenPGP protocol is to ensure the Supplier Download File is stored at rest on the suppliers computing system in a secure manner and to allow the supplier a means of verifying the Supplier Download File legitimately came from GM.

With OpenPGP, two parties who wish to communicate must setup their own public/private key pairs. Private keys are known only to the entity that created it, while public keys are exchanged, and do not need to be protected –it does not matter if an adversary is able to gain a copy of the public keys.

The following section describes how to create, exchange, and manage OpenPGP keys, as well as how to encrypt and sign messages for secure communication with GM Vehicle Cybersecurity.

4.1.1.3.1 OpenPGP security - GUID: CYS-SCPPM13d160_113

GUID: CYS-SCPPMa0c61_93 / CR 596107 (information only)

When OpenPGP is configured according to this section, the Supplier Download File will be encrypted and signed. This gives the Supplier Download File the properties of confidentiality, integrity, and authenticity. In plain terms, this means:

- If a Supplier Download File is intercepted by an adversary, they will not be able to make sense of it –only the intended recipient will be able to decrypt it
- If an adversary tampers with a message while it is in transit between the sender and receiver, it will be evident to the receiver

The receiver of the message can be certain that it originated from the expected sender

4.1.1.3.2 How OpenPGP Works - GUID: CYS-SCPPM13d160_114

GUID: CYS-SCPPM13d160_115 / CR 596107 (information only)

This section describes how the OpenPGP standard ensures the confidentiality, integrity, and authenticity of the Supplier Download File.

Confidentiality - GUID: CYS-SCPPM13d160_116

GUID: CYS-SCPPM13d160_117 / CR 596107 (information only)

First, a new random key for a symmetric-key algorithm is created for every provided Supplier Download File; for this GM specification, AES is used for symmetric key cryptography. The Supplier Download File is encrypted using AES along with this key, and this single-use AES key is encrypted using the provided supplier's public key. This way, only the supplier can decrypt Supplier Download File, because only the supplier's private key associated with the suppliers public key can be used to successfully decrypt the Supplier Download File.

Integrity and authenticity - GUID: CYS-SCPPM13d160_118

GUID: CYS-SCPPM13d160_119 / CR 596107 (information only)

A secure hash of the Supplier Download File is computed, and it is signed using a GM private key. When the supplier receives the Supplier Download File, they are able to verify the signature using the GM public key which is provided to the supplier during the onboarding process. This allows the supplier to be certain that the hash came from GM. Finally, the supplier computes the secure hash of the received message and compares it to the signed hash received with the Supplier Download File. If they match, the supplier can be certain that the Supplier Download File came from GM, and that the Supplier Download File has not been tampered with in transit.

4.1.1.3.3 OpenPGP Software - GUID: CYS-SCPPM13d160_120

GUID: CYS-SCPPM13d160_121 / CR 596107

Any software that is compliant with RFC 4880 'OpenPGP message format' shall be used.

4.1.1.3.4 Creating and Managing OpenPGP Keys - GUID: CYS-SCPPM13d160_122

GUID: CYS-SCPPM13d160_123 / CR 596107 (information only)

OpenPGP security hinges entirely on the assumption that only the party that creates a public/private key pair has knowledge of that private key. This section deals with the creation of these key-pairs, and how they are exchanged.

Key Generation - GUID: CYS-SCPPM13d160_124

GUID: CYS-SCPPM13d160_125 / CR 596107

An OpenPGP key-pair shall be generated using the following settings:

- Signing algorithm: RSA
- Encryption algorithm: RSA
- Key usage: Encryption, signing, authentication, certification
- Key length: 2048 bits
- Sub-keys are not used
- Keys do not expire
- Algorithm preferences: AES128, 3DES, SHA256, SHA1, Uncompressed

GUID: CYS-SCPPM13d160_126 / CR 596107

While the 3DES and SHA1 algorithms must be supported as per the OpenPGP standard, they shall not be used for securely exchanging material with GM. AES128 and SHA256 shall be used instead.

GUID: CYS-SCPPM13d160_127 / CR 596107 (information only)

It may take several minutes to generate a key pair. This is normal, and it is because it can take a while to generate sufficient entropy for the keys. Actively using the computer while keys are being generated may speed this up because user actions can help create additional entropy.

GUID: CYS-SCPPM13d160_128 / CR 596107

The 'real name' for the OpenPGP key pair shall be the requesting employee's full name.

GUID: CYS-SCPPM13d160_129 / CR 596107

The 'e-mail' for the OpenPGP key pair shall be the requesting employee's company e-mail address.

GUID: CYS-SCPPM13d160_130 / CR 596107

The OpenPGP key pair 'comment' shall be left blank.

GUID: CYS-SCPPM13d160_131 / CR 596107 (information only)

GNU Privacy Guard command for generating OpenPGP key pair

```
$> gpg --batch --gen-key keygen_params
```

Contents of 'keygen_params' file

```
Key-Type: 1
Key-Length: 2048
Key-Usage: encrypt, sign, auth
Name-Real: employee_name
Name-Comment:
Name-Email: employee.email@company.com
Expire-Date: 0
Preferences: AES128 3DES SHA256 SHA1 Uncompressed
%pubring public_key
%secring private_key
%commit
%echo done
```

GNU Privacy Guard command to add GM's public key to key ring

```
$> gpg --import public.key
```

GNU Privacy Guard command to verify and decrypt

```
$> gpg --decrypt received_file_from_GM > decrypted_file
```

[Storing and Backing Up Keys](#) - GUID: CYS-SCPPM13d160_132

GUID: CYS-SCPPM13d160_133 / CR 596107

The supplier shall be responsible for securely storing and backing up their OpenPGP key-pair.

[Compromised Keys](#) - GUID: CYS-SCPPM13d160_134

GUID: CYS-SCPPM13d160_135 / CR 596107

In the event that the supplier has reason to believe their OpenPGP private key may have been compromised, the supplier shall immediately notify the security admin.

[Exchanging Public Keys](#) - GUID: CYS-SCPPM13d160_136

GUID: CYS-SCPPM13d160_137 / CR 596107 (information only)

The OpenPGP standard refers to ASCII encoding as 'ASCII armored', which is type of Base64 encoding.

GUID: CYS-SCPPM13d160_138 / CR 596107 (information only)

GM will provide its ASCII-armored public key to the supplier during the onboarding process.

GUID: CYS-SCPPM13d160_139 / CR 596107

OpenPGP public key fingerprints shall be confirmed over the phone before keys are trusted.

GUID: CYS-SCPPM13d160_140 / CR 596107

The supplier shall provide their ASCII-armored public key to GM during the onboarding process.

GUID: CYS-SCPPM13d160_141 / CR 596107

Private keys shall never be communicated over e-mail to anyone.

4.1.1.3.5 Receiving Data From GM - GUID: CYS-SCPPM13d160_142

GUID: CYS-SCPPM13d160_143 / CR 596107

All Supplier Download Files received from GM shall be verified by the supplier with GM's public key, and decrypted with the supplier's private key.

GUID: CYS-SCPPM13d160_144 / CR 596107

Any message which fails signature verification shall be discarded and communicated to the Security Administrator.

4.2 Verification Data Interface - GUID: CYS-SCPPM893c3_63

GUID: CYS-SCPPMa0c61_99

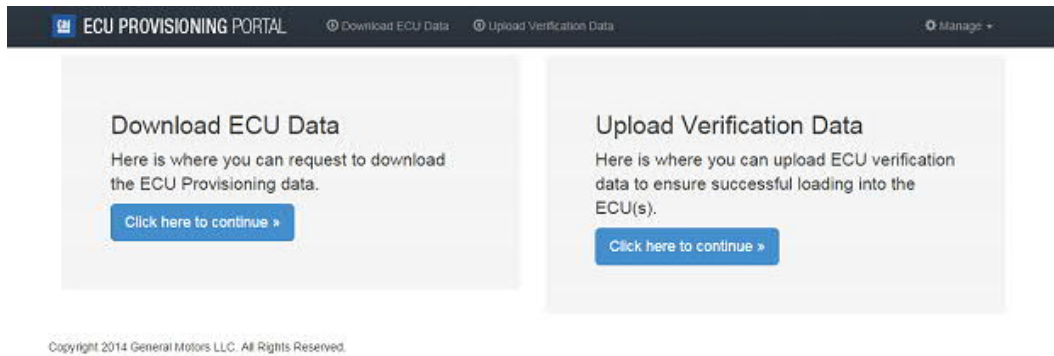
The Verification Data Interface requirements are only applicable to designs which support a security peripheral.

4.2.1 Verification Data Interface Input - GUID: CYS-SCPPM13d160_145

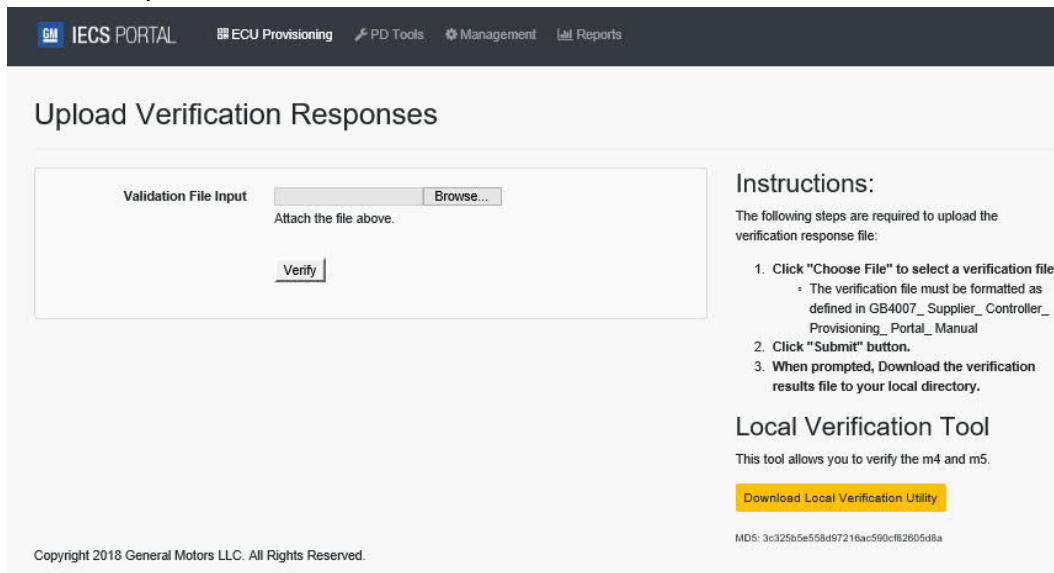
GUID: CYS-SCPPM893c3_64 / CR 596107 (information only)

The supplier will be responsible to provide the security peripheral generated M4 and M5 messages resulting from loading of the MASTER_ECU_KEY and the UNLOCK_ECU_KEY to GM through the verification data interface. The supplier may provide this data in a batch form or on an individual ECU basis. The Verification Data Interface accepts a file known as a Supplier Verification File which is in the XML format. This file must contain the ECU_ID provisioned in the ECU, ECU Serial Number, Verification Messages M4 and M5 resulting from loading of MASTER_ECU_KEY and Verification Messages M4 and M5 resulting from loading of the UNLOCK_ECU_KEY for each device manufactured.

Once logged into the ECU Provisioning Portal, the following screen will be presented. For purposes of uploading the Supplier Verification File, the user will need to click the "Upload Verification Data" button shown on the right pane of the screenshot below.



After selecting the “Upload Verification Data” button, the following screen will be presented. From this screen, the user will choose the Supplier Verification File which is locally stored on the user’s computer and then select the “submit” button.



GUID: CYS-SCPPM893c3_66 (parameter definition)

The ECU_ID shall be included in the verification file as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM893c3_65 / CR 1007259 (parameter definition)

The ECU Serial Number along with trace data values of that ECU, which is defined by GMW15862,

shall be included in the Local Supplier Verification File and can be up to 64 ASCII characters.

GUID: CYS-SCPPM893c3_67 (parameter definition)

The M4 verification message resulting from loading the MASTER_ECU_KEY shall be included in the verification file as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM893c3_68 (parameter definition)

The M5 verification message resulting from loading the MASTER_ECU_KEY shall be included in the verification file as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM893c3_69 (parameter definition)

The M4 verification message resulting from loading the UNLOCK_ECU_KEY shall be included in the verification file as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM893c3_70 (parameter definition)

The M5 verification message resulting from loading the UNLOCK_ECU_KEY shall be included in the verification file as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPMa0c61_98 (parameter definition)

The XML format of the Supplier Verification File shall be as follows:

```
<ecuRecordList>
  <ecuRecord>
    <ecuid>INFO</ecuid>
    <serialNo>INFO</serialNo>
    <mkm4>INFO</mkm4>
    <mkm5>INFO</mkm5>
    <ukm4>INFO</ukm4>
    <ukm5>INFO</ukm5>
  </ecuRecord>
  <ecuRecord>
    <ecuid>INFO</ecuid>
    <serialNo>INFO</serialNo>
    <mkm4>INFO</mkm4>
    <mkm5>INFO</mkm5>
    <ukm4>INFO</ukm4>
    <ukm5>INFO</ukm5>
  </ecuRecord>
</ecuRecordList>
```

File element description:

<ecuRecordList>	→ This tag designates the start of a Supplier Verification File
<ecuRecord>	→ This tag designates the start of a separate ECU
<serialNo>	→ This tag designates the start of the Serial Number
<mkm4>	→ This tag designates the start of the master key M4
<mkm5>	→ This tag designates the start of the master key M5
<ukm4>	→ This tag designates the start of the unlock key M4
<ukm5>	→ This tag designates the start of the unlock key M5

INFO is replaced with the proper value corresponding to the opening/closing tags within the element. INFO within the XML file must be the Base 64 encoded representation of the binary data obtained from the ECU (BINARY – BASE 64 (ENCODE)).

GUID: CYS-SCPPM13d160_146 / CR 596107 (information only)

The name of the uploaded file is not defined by GM and is left to the discretion of the ECU supplier.

4.2.2 Verification Data Interface Output - GUID: CYS-SCPPM13d160_147

GUID: CYS-SCPPM13d160_148 / CR 596107 (information only)

The Portal will process the verification data included in the Supplier Verification File and provides an xml file containing the verification results. The Portal includes a summary message indicating the overall status of the request as shown in the following screen shot (circled in red). This summary includes the number of ECU Provisioning sets processed, the number of ECU Provisioning sets which failed verification, and the number of ECU Provisioning sets which passed verification. ECUs which failed the verification process must not be provided to GM or related service shops. Refer to GUID CYS-SCPPMb4d180_245 for more details on the failed ECUs.



GUID: CYS-SCPPM13d160_149 / CR 2435625 (parameter definition)

The output file is in the xml format which contains the following elements:

```
<validationRecords>
  <validationRecord>
    <ecuid>INFO</ecuid>
    <mkValid>INFO</mkValid>
    <ukValid>INFO</ukValid>
  </validationRecord>
</validationRecords>
```

File element description:

<validationRecords>

This tag designates the start of a verification summary output file

<validationRecord>

This tag designates the start of ECU specific verification information

<ecuid>

This tag corresponds to the data representing the ECU_ID of the ECU processed for verification. The data corresponding to this tag is represented as a big endian, Base64 encoded representation of the 16 byte binary ECU_ID.

<MkValid>

This tag corresponds to the data representing verification status of the ECU's MASTER_ECU_KEY. The data corresponding to this tag will either be TRUE or FALSE where TRUE represents a successful verification of the M4 and M5 messages.

<UkValid>

This corresponds to the data representing verification status of the ECU's UNLOCK_ECU_KEY. The data corresponding to this tag will either be TRUE or FALSE where TRUE represents a successful verification of the M4 and M5 messages.

GUID: CYS-SCPPM13d160_150 / CR 596107 (information only)

The output file will have a naming convention consistent with "KeyVerOutput-<YYYY-MM-DD-HH-MM-SS>-mkuk1_5.xml".

4.3 ROW ECU Provisioning Portal Registration - GUID: CYS-SCPPM334b60_204

GUID: CYS-SCPPMe5f9c0_277 / CR 2096644

All GM Suppliers shall register into GM Supply power access web site: <https://www.gmsupplypower.com>.

Upon registering, each user gets their own unique GMIN.

Information: If you need any assistance, refer to the instructions specified in website "GM Supply Power Help"

GUID: CYS-SCPPMe5f9c0_278 / CR 2096644

After registering in GM supply power access, all suppliers shall register their modules in ECU Provisioning portal.

ECU Provisioning portal registration can be done by completing the CYS2226 - IECS Registration Form and submitting to GM Cybersecurity team. After the portal registration is complete, suppliers can then logon to GM IECS Web Portal to download the keys for the assigned ECUs.

Note: If the user already has access to ECU Provisioning portal/GM IECS Web Portal, there is no need to fill out CYS2226 - IECS Registration Form again.

Note 2: If CYS2226 - IECS Registration Form cannot be found in the ECU's Appendix F14 requirements document, contact project specific GM Product Cybersecurity Program Manager for CYS2226 - IECS Registration Form.

4.4 China GEM ECU Provisioning Portal Registration - GUID: CYS-SCPPMe5f9c0_279

GUID: CYS-SCPPMe5f9c0_280 / CR 2096644

China GEM program (Vehicles sold in China) suppliers should contact their Cybersecurity PM to get the links for ECU Provisioning Portal.

5 Local Verification Utility - GUID: CYS-SCPPM13d160_151

GUID: CYS-SCPPM13d160_152 / CR 596107 (information only)

Similar to the online verification procedure described under CYS-SCPPM893c3_63, the supplier will be responsible for obtaining an initial Verification Status on the security peripheral generated M4 and M5 messages resulting from loading of the MASTER_ECU_KEY and the UNLOCK_ECU_KEY via the Local Verification Utility. The supplier may provide this data in a batch form or on an individual ECU basis. The Local Verification Utility accepts a file known as a Local Supplier Verification File which is in the XML format. This file must contain the following elements for each ECU provisioned with an MASTER_ECU_KEY and an UNLOCK_ECU_KEY:

- ECU_ID provisioned in the ECU
- ECU Serial Number
- M1, M2, and M3 messages passed to the security peripheral for provisioning of the MASTER_ECU_KEY
- M4 and M5 Verification Messages resulting from loading of MASTER_ECU_KEY
- M1, M2, and M3 messages passed to the security peripheral for provisioning of the UNLOCK_ECU_KEY

M4 and M5 Verification Messages resulting from loading of UNLOCK_ECU_KEY

GUID: CYS-SCPPM13d160_153 / CR 596107

The supplier shall use the Local Verification Utility to verify each set of M4 and M5 verification messages returned from the security peripheral as a result of provisioning the MASTER_ECU_KEY and UNLOCK_ECU_KEY.

GUID: CYS-SCPPM46b9e0_210 / CR 1007259 (information only)

There are some following gaps between verification performed by Local Verification Tool and Online verification:

- Usage of repeated ECU IDs is not checked by Local verification Tool
- Local verification cannot make sure the correct association between ECU ID and the secret keys

5.1 Local Verification Utility Input - GUID: CYS-SCPPM13d160_155

GUID: CYS-SCPPM13d160_154 / CR 596107

The ECU_ID shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_156 / CR 596107

The ECU Serial Number along with trace data values of that ECU, which is defined by GMW15862,

shall be included in the Local Supplier Verification File and can be up to 64 ASCII characters.

GUID: CYS-SCPPM13d160_157 / CR 596107

The M1 message used to provision the MASTER_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_158 / CR 596107

The M2 message used to provision the MASTER_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM13d160_159 / CR 596107

The M3 message used to provision the MASTER_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_160 / CR 596107

The M4 verification message resulting from loading the MASTER_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM13d160_161 / CR 596107

The M5 verification message resulting from loading the MASTER_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_162 / CR 596107

The M1 message used to provision the UNLOCK_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_163 / CR 596107

The M2 message used to provision the UNLOCK_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM13d160_164 / CR 596107

The M3 message used to provision the UNLOCK_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_165 / CR 596107

The M4 verification message resulting from loading the UNLOCK_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 32 byte binary value.

GUID: CYS-SCPPM13d160_166 / CR 596107

The M5 verification message resulting from loading the UNLOCK_ECU_KEY shall be included in the Local Supplier Verification File as a big endian, Base64 encoded representation of the 16 byte binary value.

GUID: CYS-SCPPM13d160_167 / CR 596107 (parameter definition)

The XML format of the Local Supplier Verification File shall be as follows:

```
<ecuRecordList>
  <ecuRecord>
    <ecuid>INFO</ecuid>
```

```

    <serialNo>INFO</serialNo>
    <mkm1>INFO</mkm1>
    <mkm2>INFO</mkm2>
    <mkm3>INFO</mkm3>
    <mkm4>INFO</mkm4>
    <mkm5>INFO</mkm5>
    <ukm1>INFO</ukm1>
    <ukm2>INFO</ukm2>
    <ukm3>INFO</ukm3>
    <ukm4>INFO</ukm4>
    <ukm5>INFO</ukm5>
  </ecuRecord>
  <ecuRecord>
    <ecuid>INFO</ecuid>
    <serialNo>INFO</serialNo>
    <mkm1>INFO</mkm1>
    <mkm2>INFO</mkm2>
    <mkm3>INFO</mkm3>
    <mkm4>INFO</mkm4>
    <mkm5>INFO</mkm5>
    <ukm1>INFO</ukm1>
    <ukm2>INFO</ukm2>
    <ukm3>INFO</ukm3>
    <ukm4>INFO</ukm4>
    <ukm5>INFO</ukm5>
  </ecuRecord>
</ecuRecordList>

```

File element description:

<ecuRecordList>	→ This tag designates the start of a Supplier Verification File
<ecuRecord>	→ This tag designates the start of a separate ECU
<serialNo>	→ This tag designates the start of the Serial Number
<mkm1>	→ This tag designates the start of the master key M1
<mkm2>	→ This tag designates the start of the master key M2
<mkm3>	→ This tag designates the start of the master key M3
<mkm4>	→ This tag designates the start of the master key M4
<mkm5>	→ This tag designates the start of the master key M5
<ukm1>	→ This tag designates the start of the unlock key M1
<ukm2>	→ This tag designates the start of the unlock key M2
<ukm3>	→ This tag designates the start of the unlock key M3
<ukm4>	→ This tag designates the start of the unlock key M4
<ukm5>	→ This tag designates the start of the unlock key M5

INFO is replaced with the proper value corresponding to the opening/closing tags within the element. INFO within the XML file must be the Base 64 encoded representation of the binary data obtained from the ECU (BINARY – BASE 64 (ENCODE)).

GUID: CYS-SCPPM13d160_168 / CR 596107 (information only)

The name of the uploaded file is not defined by GM and is left to the discretion of the ECU supplier.

5.2 Executing the Local Verification Utility - GUID: CYS-SCPPM13d160_169

GUID: CYS-SCPPM13d160_170 / CR 596107 (information only)

The Local Verification Utility (IecsM4M5.jar) is a java based application which is made available for download directly from the Supplier ECU Provisioning Portal Web Page.

GUID: CYS-SCPPM13d160_171 / CR 596107 (information only)

The following steps must be followed for execution of the Local Verification Utility:

1. Unzip the contents of the Local Verification Utility package to a selected folder on your computer
2. Save the Local Supplier Verification File to the same folder on your computer where the Local Verification Utility was unzipped. The Local Supplier Verification File should be at the same level as the Local Verification Utility .jar file within your folder hierarchy.
3. Open a command prompt by clicking 'start' and searching for cmd. Select cmd.exe.
4. Within the command prompt window, navigate to the directory where the Local Verification Utility .jar file resides (e.g. cd C:\NewFolder)
5. Type the following in the command prompt: java -jar <filename>.jar "argument" (e.g. java -jar IecsM4M5.jar "C:\NewFolder\LocalSupplierVerificationFile").
 - a. This jar file takes one argument, a file path containing the Local Supplier Verification File.
 - b. Be sure to use double quotes when surrounding the argument.
 - c. The output file containing verification results is saved to the location where the Local Verification Utility .jar file resides.

5.3 Local Verification Utility Output - GUID: CYS-SCPPM13d160_172

GUID: CYS-SCPPM13d160_173 / CR 596107 (parameter definition)

The Local Verification Utility will provide an output file describing the verification status of each ECU. The output file is in the xml format which contains the following elements:

```
<VerificationSummary>
  <totalRecords>INFO</totalRecords>
  <validRecords>INFO</validRecords>
  <invalidRecords>INFO</invalidRecords>
  <VerificationResult>
    <ecuid>INFO</ecuid>
    <MkValid>INFO</MkValid>
```

```

        <UkValid>INFO</UkValid>
    </VerificationResult>
</VerificationSummary>

```

File element description:

<VerificationSummary>

This tag designates the start of a verification summary output file

<totalRecords>

This tag corresponds to the data representing the total number of ECUs processed for verification results. The data corresponding to this tag is represented as a numeric number.

<validRecords>

This tag corresponds to the data representing the total number of ECUs which resulted in successful verification. The data corresponding to this tag is represented as a numeric number.

<invalidRecords>

This tag corresponds to the data representing the total number of ECUs which did not result in successful verification. The data corresponding to this tag is represented as a numeric number.

<VerificationResult>

This tag designates the start of ECU specific verification information.

<ecuid>

This tag corresponds to the data representing the ECU_ID of the ECU processed for verification. The data corresponding to this tag is represented as a big endian, Base64 encoded representation of the 16 byte binary ECU_ID.

<MkValid>

This tag corresponds to the data representing verification status of the ECU's MASTER_ECU_KEY. The data corresponding to this tag will either be TRUE or FALSE where TRUE represents a successful verification of the M4 and M5 messages.

<UkValid>

This corresponds to the data representing verification status of the ECU's UNLOCK_ECU_KEY. The data corresponding to this tag will either be TRUE or FALSE where TRUE represents a successful verification of the M4 and M5 messages.

</VerificationResult>

This tag designates the end of ECU specific verification information.

</VerificationSummary>

This tag designates the end of a verification summary output file.

GUID: CYS-SCPPM13d160_174 / CR 596107 (information only)

The output file will have a naming convention consistent with “LocalKeyVerOutput-yyyy-mm-dd-hh-mm-ss-<input_file_name>.xml”, where <input_file_name> is the name of the Local Supplier Verification File passed into the Local Verification Utility.

GUID: CYS-SCPPM13d160_175 / CR 2096644

The supplier shall only ship ECUs to GM and service related facilities which have received a successful verification status from the Online Verification. Refer to GUID CYS-SCPPMb4d180_245 for more details on the failed ECUs.

6 Appendix A - Provisioning requirements - GUID: CYS-SCPPM4c38a0_211

GUID: CYS-SCPPMe5f9c0_282 / CR 2096644

All Controllers with an integrated security peripheral and supporting Message Authentication delivered to GM for development and vehicle manufacturing purposes shall have each Key_Slot_Provision_State_Flag (KSPS)

shall be set equal to a value of 0x00.

Note: Refer to GB6001_Diagnostic_Infrastructure_Specification for definition of KSPS.

GUID: CYS-SCPPMe5f9c0_283 / CR 2096644

All Controllers with an integrated security peripheral and supporting Message Authentication delivered to GM for service (service parts) shall have each KSPS shall be set equal to a value of 0x01.

Note: Refer to GB6001_Diagnostic_Infrastructure_Specification for definition of KSPS.

GUID: CYS-SCPPM4c38a0_212 / CR 1615072

Following information should be used during key provisioning:

Secret Keys and Associated Controllers	Supplier Secret Key Provisioning Location			Secret Key Value		
	With an Integrated Security Peripheral		Without an Integrated Security	Value for Development Parts	Value for Production Parts	Value for Service Parts
	Key Slot Number (Dec)	Key_ID (Hex)		Refer GUIDs	Refer GUIDs	Refer GUIDs
MASTER_ECU_KEY (All Controllers)	255	0x1	Not Applicable	CYS-SCPPM893c3_40	CYS-SCPPM893c3_40	CYS-SCPPM893c3_40
UNLOCK_ECU_KEY (All Controllers)	1	0x4	Secure Static Memory	CYS-SCPPM893c3_38 / CYS-SCPPM893c3_35	CYS-SCPPM893c3_38 / CYS-SCPPM893c3_35	CYS-SCPPM893c3_38 / CYS-SCPPM893c3_35
Key_11, Key_21, Key_31, Key_41	11, 21, 31, 41	0x4	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_2, Key_12, Key_22, Key_32, Key_42	2, 12, 22, 32, 42	0x5	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_3, Key_13, Key_23, Key_33, Key_43	3, 13, 23, 33, 43	0x6	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_4, Key_14, Key_24, Key_34, Key_44	4, 14, 24, 34, 44	0x7	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_5, Key_15, Key_25, Key_35, Key_45	5, 15, 25, 35, 45	0x8	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_6, Key_16, Key_26, Key_36, Key_46	6, 16, 26, 36, 46	0x9	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_7, Key_17, Key_27, Key_37, Key_47	7, 17, 27, 37, 47	0xA	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_8, Key_18, Key_28, Key_38, Key_48	8, 18, 28, 38, 48	0xB	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_9, Key_19, Key_29, Key_39, Key_49	9, 19, 29, 39, 49	0xC	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220
KEY_10, Key_20, Key_30, Key_40, Key_50	10, 20, 30, 40, 50	0xD	Not Applicable	CYS- SCPPM4c38a0_213	CYS- SCPPM4c38a0_213	CYS-SCPPM4c38a0_220

GUID: CYS-SCPPM4c38a0_213 / CR 2096644

All Controllers with an integrated security peripheral and supporting Message Authentication delivered to GM for development and vehicle manufacturing purposes shall have security peripheral memory slots KEY_2 thru KEY_<n> (where n = NOKSS) set equal to a valid 128-bit value of 0 (zero).

GUID: CYS-SCPPM4c38a0_220 / CR 2096644

All Controllers with an integrated security peripheral delivered to GM for service (service parts) shall have security peripheral memory slots KEY_2 thru KEY_<n> (where n = NOKSS) provisioned with 128 bit unique random numbers.

GUID: CYS-SCPPMb4bfc0_221 / CR 2096644

All Controllers with an integrated security peripheral not supporting message authentication, shall have security peripheral memory slots KEY_2 thru KEY_<n> (where n = NOKSS) provisioned with 128 bit unique random numbers for all use cases like Development, Service and Manufacturing parts.

GUID: CYS-SCPPMe5f9c0_284 / CR 2096644

All Controllers with an integrated security peripheral not supporting message authentication, shall have each Key_Slot_Provision_State_Flag set equal to a value of 0x01 for all use cases like Development, Service and Manufacturing parts.

Note: Refer to GB6001_Diagnostic_Infrastructure_Specification for definition of Key_Slot_Provision_State_Flag.

GUID: CYS-SCPPM4c38a0_219 / CR 1615072

The tool generating random numbers shall meet requirements as specified by CYS9000 - Random_Number_Generator_Security.

GUID: CYS-SCPPM4c38a0_218 / CR 2096644

When required to load Keys into security peripheral slots KEY_2 thru KEY_<n> (where n = NOKSS), the security flags (F_ID) loaded into the key slot shall be set such that the slot is not Write Protected (WRITE_PROTECTION = 0).

GUID: CYS-SCPPM4c38a0_217 / CR 2096644

When required to load Keys into security peripheral slots KEY_2 thru KEY_<n> (where n = NOKSS), the security flags (F_ID) loaded into the key slot shall be set such that the slot is not disabled on boot failure (BOOT_PROTECTION = 0).

GUID: CYS-SCPPM4c38a0_216 / CR 2096644

When required to load Keys into security peripheral slots KEY_2 thru KEY_<n> (where n = NOKSS), the security flags (F_ID) loaded into the key slot shall be set such that the slot is enabled on debugger activation (DEBUGGER_PROTECTION = 0).

GUID: CYS-SCPPM4c38a0_215 / CR 2096644

When required to load Keys into security peripheral slots KEY_2 thru KEY_<n> (where n = NOKSS), the security flags (F_ID) loaded into the key slot shall be set such that the slot can be updated without a valid UID (WILDCARD = 0).

GUID: CYS-SCPPM4c38a0_214 / CR 2096644

When required to load Keys into security peripheral slots KEY_2 thru KEY_<n> (where n = NOKSS), the security flags (F_ID) loaded into the key slot shall be set such that the slot can be used for MAC generation and verification (KEY_USAGE = 1 and VERIFY_ONLY = 0).

GUID: CYS-SCPPMb4bfc0_222 / CR 2096644

When generating M1, M2, and M3 messages for purposes of loading security peripheral key slots KEY_2 thru KEY_<n> (where n = NOKSS), a counter value (i.e. C-ID) greater than 100 shall not be used.

GUID: CYS-SCPPMe5f9c0_285 / CR 2096644

After provisioning M1, M2 and M3 for a particular key slot, the resultant M4/M5 generated from the module shall be checked against M4/M5 generated while creating M1, M2 and M3 for that key slot.

GUID: CYS-SCPPMe5f9c0_287 / CR 2096644

All controllers with an integrated security peripheral delivered to GM development and vehicle manufacturing purposes shall use the following test vectors for KEY_2 to KEY_<n> (where n = NOKSS).

Keyslot #	M1,M2 and M3 values
2	M1 = 00000000000000000000000000000055 M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = 1087bc086f56b60548fdf7359eeb1f19
3	M1 = 00000000000000000000000000000066 M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = a9184c8d820ca2e393386b16e5c2380d
4	M1 = 00000000000000000000000000000077 M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = 00d5ca7e375c79fe2e91dd9fa20dc9c2
5	M1 = 00000000000000000000000000000088 M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = f0c1b56e23459ab9e7736859211ac24f
6	M1 = 00000000000000000000000000000099 M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = d427f84bc02aed37934e74267c61334f
7	M1 = 000000000000000000000000000000aa M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = be714f0678cc40aa792295028eb4969b
8	M1 = 000000000000000000000000000000bb M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = 7bf9328636d3d3e795b1b0e468186e4c

9	M1 = 0000000000000000000000000000cc M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = d326b4e90efbf17bccf67714867b29c2
10	M1 = 0000000000000000000000000000dd M2 = cb658ac715cf5c8a517c6f8939e681d2c2b9902502304dc4f058d4b68d106218 M3 = 66b9ec2cc8678640ee06d88cbee69b07
11	M1 = 000000000000000000000000000044 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 252cd874b0ebd9d6b4c7ccac59451fbd
12	M1 = 000000000000000000000000000055 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 0ae99280ddeccc0b55ddb94e4a856628
13	M1 = 000000000000000000000000000066 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 56ab889fce69d9ab8db27477827e2311
14	M1 = 000000000000000000000000000077 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 6d9c490230d2db2e0b436114a5c45162
15	M1 = 000000000000000000000000000088 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 2761f40731dd66801ed7cea13fa69362
16	M1 = 000000000000000000000000000099 M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 6da5bfdcf70c7389e1c3bfcdfaa9ef0a
17	M1 = 0000000000000000000000000000aa M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = ec20b2a974091b6481a9451bda4d13a5
18	M1 = 0000000000000000000000000000bb M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = c51b3e59ae5093283c86ab0cc2becff1
19	M1 = 0000000000000000000000000000cc M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = 254f1d2c051c5ff5de18774a4d3552fd
20	M1 = 0000000000000000000000000000dd M2 = 01e0c13c47620fa8263171fae3c7e24f0869457cac242fdf269ca3420c5382b1 M3 = e1dda92aa9c3aa6f4be954692048bb72
21	M1 = 000000000000000000000000000044 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = f242cb8a0706b9fc8383c86ef6459f05
22	M1 = 000000000000000000000000000055 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 66d89f9b2d8646af55a36055bb13aa08

23	M1 = 00000000000000000000000000000066 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 7eb1ebdbc4d81ddb7719588905ebd397
24	M1 = 00000000000000000000000000000077 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 59fc5785370b124bc404c41f3fd91de9
25	M1 = 00000000000000000000000000000088 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 60d36bbe6fd13988bfaf75484c5bcb5d
26	M1 = 00000000000000000000000000000099 M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 729e39362019e588851d734900a0624a
27	M1 = 000000000000000000000000000000aa M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = c24ec4a2b52b0335db5688f8eb084db0
28	M1 = 000000000000000000000000000000bb M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 8b0b2d041aa92a3a79b96960907e86aa
29	M1 = 000000000000000000000000000000cc M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = f0682b3ac70a608b5ca3955471c3692e
30	M1 = 000000000000000000000000000000dd M2 = 0d33859a7450c02c601b28553efa0c8a01b0431e38e0b39d1a526af0a3afc720 M3 = 93c43cde6a33553629b8488a7d24cd60
31	M1 = 00000000000000000000000000000044 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = f2189e39fa8709b1d9f24a261a8f9066
32	M1 = 00000000000000000000000000000055 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = 150dfcfd3f921ee8e5093c83a28fa5d8
33	M1 = 00000000000000000000000000000066 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = b24c7f3dc8e0ef5da6f38dd795172d62
34	M1 = 00000000000000000000000000000077 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = 36141074ed3e039c1e9d671a128d15b1
35	M1 = 00000000000000000000000000000088 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = f025314ddb665056c88594b3c445b729
36	M1 = 00000000000000000000000000000099 M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = b513bc38992aed6b8e63ef02666459d7

37	M1 = 0000000000000000000000000000aa M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = a449f04fbc70085a3fe06bd446422dbf
38	M1 = 0000000000000000000000000000bb M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = df201f01e880c03b0c5976f5eed6e650
39	M1 = 0000000000000000000000000000cc M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = b313c486642a13fb0508fe5492f69f4a
40	M1 = 0000000000000000000000000000dd M2 = 22668cb56bec7bf964285c97cf849e92df15d1ec7b9e39bc1eb31536871fbb8a M3 = e9734ca251ad7d69ac99bd8e89f0b695
41	M1 = 000000000000000000000000000044 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 7a383639d6faef8a91949d9dbb9fedcb
42	M1 = 000000000000000000000000000055 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 5256835b256eca993547ba289d120b70
43	M1 = 000000000000000000000000000066 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = a8f57e6ca204f88540643508dc0e6e6b
44	M1 = 000000000000000000000000000077 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = fbcaa5021d30a1f44133033a5840aa99
45	M1 = 000000000000000000000000000088 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 6395529081938460c35b90dd2c0afeaf
46	M1 = 000000000000000000000000000099 M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = aa87b7dafb122a3fdcf712ff756ac77a
47	M1 = 0000000000000000000000000000aa M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 411871f78d5352c82f41a01f3f5fc0bf
48	M1 = 0000000000000000000000000000bb M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = b1f8502a87d2bb5397a52ea6c82fcf1
49	M1 = 0000000000000000000000000000cc M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 091f4b282356a1625abcd7e6f1a8610
50	M1 = 0000000000000000000000000000dd M2 = 52ff88157e69444c71663f0226d03379b3f213f2b99a2e95f5b1f56db139c6af M3 = 857ac48207774fb3a05d8e351f9c2de1

Note: Above test vectors are based on the key values of zero, CID =90 and FID =4. In case of an issue using these test vectors, contact GM Cybersecurity Program Manager.

GUID: CYS-SCPPMe5f9c0_312 / CR 2096644

Parts that fail testing in CYS-SCPPMd7ef40_303 shall not be delivered to GM.

GUID: CYS-SCPPMe5f9c0_313 / CR 2096644

DID F081 response shall be checked after provisioning corresponding delivery parts as mentioned in GUID CYS-SCPPMd7ef40_297 and CYS-SCPPMd7ef40_299

Note 1: Successful key provisioning through RID 200 will update the corresponding KSPS from 0 to 1.

Note 2: Successful key provisioning through RID 272 will not update KSPS

GUID: CYS-SCPPMe5f9c0_314 / CR 2096644 (information only)

Suppliers are free to choose and use any desired method to provision keys into the controllers during ECU mfg. If they intend to use GM defined DID/RIDs for this purpose, following is the suggested high level procedure:

1. Provision ECU_ID via JTAG
2. Provision Master Key via RID 272
 - a. Collect M4/M5 data returned back
3. Provision Unlock Key via RID 272
 - a. Collect M4/M5 data returned back
4. Verify M4/M5 for Master and unlock keys with the GM server per GB4007.
5. For parts provided to GM Manufacturing and development, provision zero keys for the Key Slots 2 through 20 via RID 272 (use CID value less than 10 when generating M1, M2 and M3)
 - a. Verify M4/M5 with the M4/M5 generated by the tool generating M1-M3 for the keys for each key slot.
6. For parts provided to Service (service parts), provision random keys for the Key Slots 2 through 20 via RID 200
 - a. Verify M4/M5 with the M4/M5 generated by the tool generating M1-M3 for the keys for each key slot.
7. For parts provided to GM Manufacturing and development, provision keys defined via CYS-SCPPMd7ef40_303 using RID 272.
8. Verify DID F081 response per CYS-SCPPMd7ef40_297 and CYS-SCPPMd7ef40_299

Production and Service parts are not interchangeable. i.e. Production parts cannot be used in service and service parts cannot be used in GM development/production. Following table gives high level information on the deliverable parameter values.

	Development/Production ECU	Service ECU	Notes
MEC	0x50	0x00	See GB6001 for more details
Key Provisioning State Flags (KSPS)	0	1	See GB4007 for more details
Key Slots 2-NOKSS	Valid key value of 0	Valid random key	See GB4007 for more details

7 Appendix B - Machine to Machine (M2M) Interface - GUID: CYS-

SCPPMe5f9c0_293

GUID: CYS-SCPPMe5f9c0_294 / CR 2096644

Suppliers shall use the M2M interface for all the IECS activities like downloading ECU_ID/keys and M4/M5 verification with IECS server. GM defined M2M interface shall be used only on supplier server communicating with the GM IECS server.

Note: The intention of this requirement is to ensure that M2M interface is not misused by installing and utilizing the interface on transferable assets (e.g. laptops, tablets, etc.).

Note 2: The automation process from supplier server to supplier tools is left to the discretion of the supplier.

7.1 Request Certificate for Supplier M2M Server - GUID: CYS-SCPPMe5f9c0_295

GUID: CYS-SCPPMe5f9c0_303 / CR 2435625 (information only)

Once the steps in this section are completed, supplier will be able to use the GM IECS certificate granted to them to install on their machine and access the IECS Supplier Direct / Machine-to-Machine (M2M) APIs.

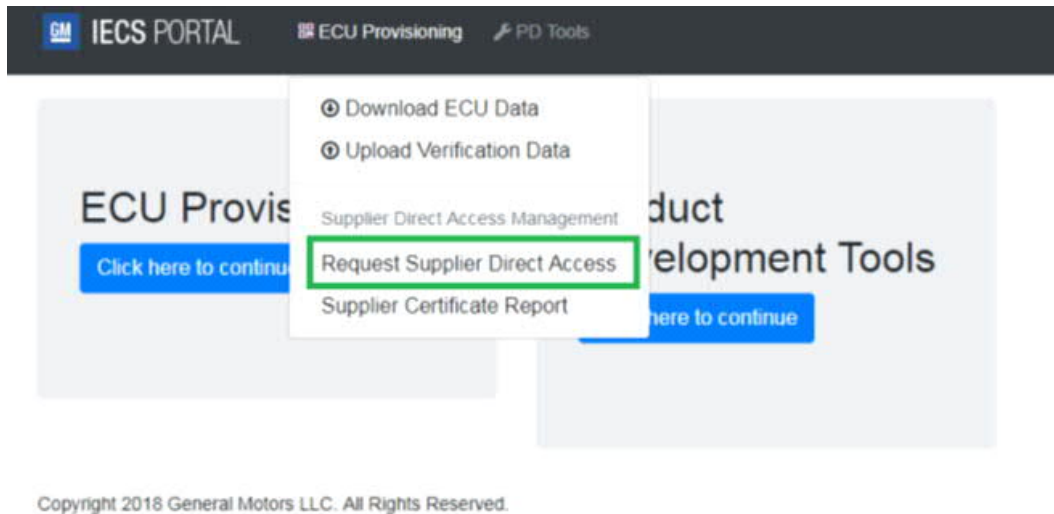
M2M registration can be done by completing the CYS2226 - IECS Registration Form and submitting to GM Cybersecurity team (cys@gm.com). After the M2M registration is complete, suppliers can then logon to GM IECS Web Portal to request M2M certificate for the assigned ECUs.

Note: Certificate will have an expiration date and users can see the expiration date in the certificate itself. Suppliers need to be aware of the expiration date and renew the certificate in advance (up to 3 months) for the GM deliverables. For more information on certificates, refer M2M User Guide.

7.1.1 Steps to Request M2M Certificate - GUID: CYS-SCPPMe5f9c0_310

GUID: CYS-SCPPMe5f9c0_311 / CR 2096644 (information only)

Log in to GM IECS Web Portal. Click on ECU Provisioning on the navigation bar and click on "Request Supplier Direct Access" as shown below.



Note: If the above options are not shown on the screen, please contact GM Cybersecurity team (cys@gm.com) as they must grant access before one can view this option.

Once on the Request Supplier Direct Access page, you will need to fill out a form. There are instructions on the right side of the page that serve as a guide for filling out the form.

Here is an example of a request:

a) Provide the Machine Hostname for the machine that will have the certificate.

The screenshot shows the 'Request Supplier Direct Access' form. The title is 'Request Supplier Direct Access' with a subtitle 'Please fill out the form below'. The form has a section for 'Machine Hostname' with a text input field containing 'Machine Using M2M nam corp.com' and a 'Next' button. To the right, there are 'Instructions' for requesting a certificate, including a list of steps and a 'Supplier Certificate Report' button. The background shows the GM IECS PORTAL navigation bar.

b) Provide the Machine's CSR. Detailed instructions on how to create a CSR can be found in M2M User Guide (M2M User Guide can be downloaded from GM IECS Web Portal).

Provide CSR

This requires the user to construct a request in the PKCS #10 format. We recommend using the [OpenSSL](#) tool to generate your CSR. After installing OpenSSL, use this command by replacing the marked fields with your own information:

```
openssl req -new -newkey rsa 2048 -nodes -out [Common Name].csr -keyout [Common Name].key -subj "/C=[Country]ST=[State]X.[o]=[Email Address]O=[org (or XM)]OU=[Department]CN=[Common Name]"
```

Instructions:

If you have already made a request for a certificate, you can view the status by clicking the button below:

Supplier Certificate Report

The following steps are required to receive a certificate for your machine to use with M2M:

1. Enter the machine's hostname that you are requesting a certificate for.
2. Enter the CSR you generated using the OpenSSL command.
3. Enter the Machine's Public Key.
4. Click "Submit" button.
5. Once your request is approved or rejected, you will be notified via email. Go to the "Supplier Certificate Report" page to view the status of your request.
 - In the email, if you are approved you should receive the certificate for your machine as an attachment; otherwise, you will receive the reason for denial.

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IECS PORTAL
ECU Provisioning
PD Tools

Request Supplier Direct Access

Please fill out the form below

Provide Machine Public Key

This requires the user to generate a PGP Key(Public Key) that will be used for requesting ECU provisioning data sets.

```

-----BEGIN PGP PUBLIC KEY BLOCK-----

mQENRfFq7eq@CADQV3M8p9ASfWQ28heqFicq8qT8pgXwYyWYQv17JuxQ3X
3ZoS2YVXG1xK7Zukn3mbv4f7hpsG8BY8rmFyc1LqaFrRZa8Mm1mYEa5GRQy
oyet812dLNTa2ehMcZy+ZasrRMe1G2Pe+gr6GzK9K9887GdaFkPY513p
1dR8m+G2M2eG45+Znk13H5M44RHeH1nPvW079277ox7P81yWpG-A2Nod1e
MX8VgmQ3Kb4Q+K16epX02C3NP+J23mM7hL3Yy8y+G8KApe416+VW92Q
Q3aRuyRTIeVAgOONB5G8bVQ8UkPG01H96IA8EBAAGORVjN2b1uQFORBMSAAJ
FfEENFR1M0a4yb6+XiamPFQV7AQIFAs7eqCQeMFCA8Bp4GFQoYCAICRRYy
AwEC+q6CF4AACq8QmPFQV7AQIKT1Gf+pgRReh8Qw78A6wW62+J8r4D6Ch7
0+JuaMpr56a4M0rZmQuKwWQOmC54TE4h0Xesspb7YReNemQdm3L3DeyXZTEC8Q
9pb8G3uQ8Q8ePFYg454cJecPFK74KQsa+lyp8K7cEgcnMmW8HAZK8yTQ
y4H8hbmFYm8Vv1+ye4eC3dC8F6AImbvJfYKFSu6Q873JuxuPyQ8h33
Xx8uUfC5LmDh8t8u8p8d8u8h8a8u8A84561VW5gR0e
W80Dm8d45up8FXT7a72PQNR8p8d8K8mR8KfC8bYQ8x1D4f8G8b8Yy406+
pSPwAFy8Q8p8Y8LH8C893n+HwGQ8Q8Q8V8H8TF8M85Y8f8L8d
XN8hamC8F40D8H5Y2C86+H281n2SQ8MP8fC8B8PC8R8C8Z8u8c8v8H8U3801vT
jA==
=KXMJ
-----END PGP PUBLIC KEY BLOCK-----

```

Previous
Submit

Instructions:

If you have already made a request for a certificate, you can view the status by clicking the button below:

[Supplier Certificate Report](#)

The following steps are required to receive a certificate for your machine to use with M2M:

1. Enter the machine's hostname that you are requesting a certificate for.
2. Enter the CSR you generated using the OpenSSL command.
3. Enter the Machine's Public Key.
4. Click "Submit" button.
5. Once your request is approved or rejected, you will be notified via email. Go to the "Supplier Certificate Report" page to view the status of your request.
 - In the email, if you are approved you should receive the certificate for your machine as an attachment, otherwise, you will receive the reason for denial.

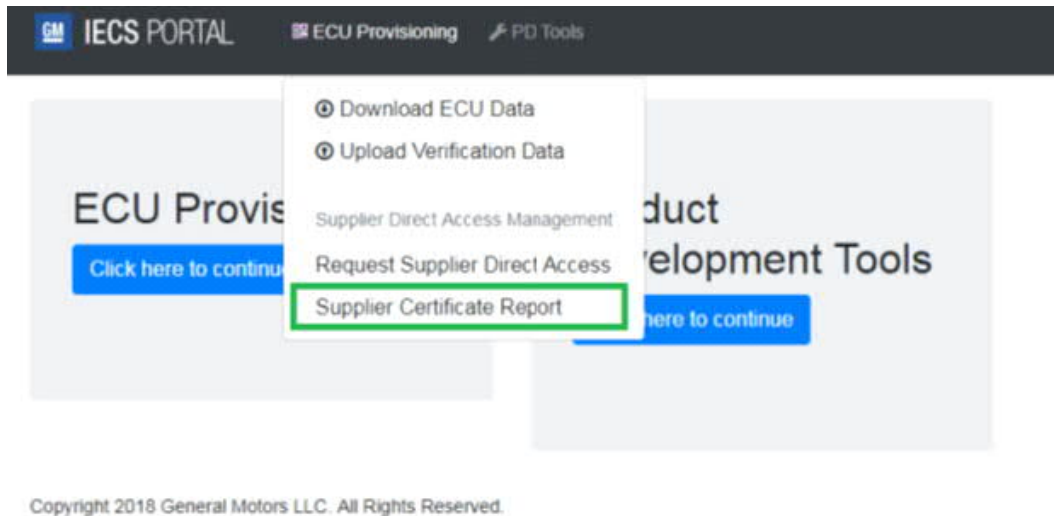
Once all the fields are filled out correctly, click the Submit button. A success alert will appear at the top of the page. An IECS Administrator will receive and review your request once it has been posted in their queue.

7.2 View Certificate Request Status - GUID: CYS-SCPPMe5f9c0_308

GUID: CYS-SCPPMe5f9c0_309 / CR 2096644 (information only)

GM IECS server has a page specifically for viewing the M2M certificate requests. To view the page, follow the steps below:

Log in to GM IECS Web Portal. Click on ECU Provisioning on the navigation bar and click on Supplier Certificate Report as shown below.



Note: If the above options are not shown on the screen after submitting IECS registration for M2M, indicates corresponding GMIN is still not approved to use M2M interface, contact GM Cybersecurity team(cys@gm.com).

After user clicks on Supplier Certificate Report , page will display a page as shown below.

The screenshot shows the 'Supplier Certificates Report' page. At the top, there are 'CSV' and 'Excel' buttons. A green arrow points to the 'Excel' button. Below the buttons is a search bar. The main part of the page is a table with the following columns: 'Common Name', 'Supplier', 'GMIN', 'Date Requested', 'Approval Status', 'Processed Date', and 'Expiration Date'. The table contains one row of data: 'commonName', 'GM', '16-Oct-2018 04:10 PM', 'PENDING', and empty cells for 'Processed Date' and 'Expiration Date'. Below the table, it says 'Showing 11 to 11 of 11 entries'. At the bottom right, there are pagination controls: 'Previous', '1', '2', '3', and 'Next'. The '3' is highlighted in blue. At the bottom left, it says 'Copyright 2018 General Motors LLC. All Rights Reserved.'

Test Application is provided along with M2M release ("M2M Utility") to make sure IECS server can be communicated with valid certificates. Just install certificate and download the application and interface API(s) from GM IECS Web portal. For more information on using the Test Application, refer M2M User Guide. "M2M Utility" can be downloaded from GM IECS Web portal location as shown below.

Local Verification Tool

This tool allows you to verify the m4 and m5.

[Download Local Verification Utility](#)

MD5: 004fe210a3a444f224e98965aad89002

M2M Application Interface

This tool automates the communication between IECS and the supplier manufacturing server.

[Download M2M Utility](#)

MD5: 96a2c6ab57ef4e15a03319699e12dcd1

8 Change History

Change request number	Date	Description of changes
2435625	2020-06-21	Requirements added for HSM implementation with external key storage. 50 key(s) test vectors are added.
2096644	2019-05-06	M2M requirements are added. Provisioning requirements clarifications are updated in Appendix A
1496442	2018-08-16	Updated IECS Web URL screens. Added process clarifications.
1007259	2016-01-14	Added ECCN number and clarifications
1022628	2016-01-14	Information about RID 0x0272 is added. Key provisioning clarifications updated in Appendix A
884785	2016-01-14	Added URL details for GM Supply Power Access and IECS Portal
596107	2014-10-23	Added definition and requirements for OpenPGP usage Added details pertaining to the Supplier Portal Web UI Added definition and requirements for Local M4/M5 Verification Utility
470147	2014-07-14	Initial release of GB4007 - Supplier Controller Provisioning Portal Manual